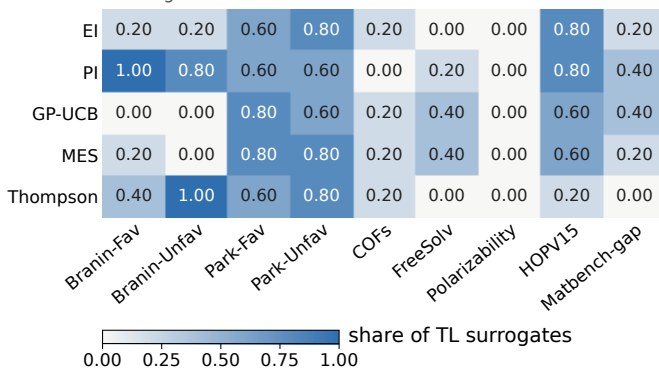
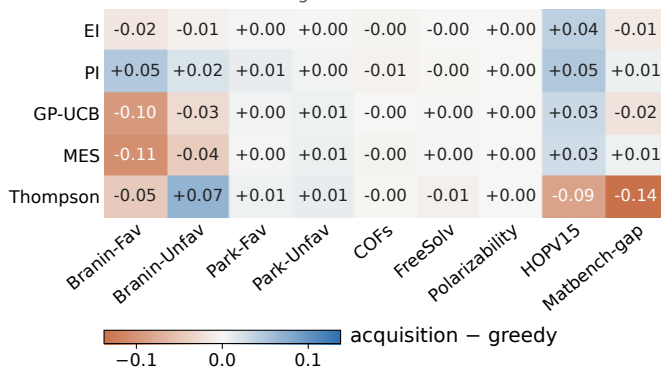


a Share of TL surrogates attaining more than greedy

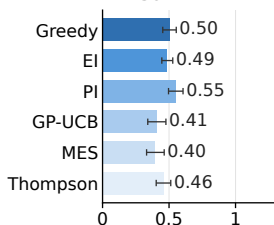
5 surrogates · 20 seeds each

**b** Mean attainment difference vs greedy

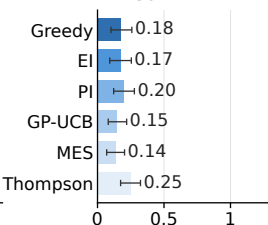
mean over the 5 surrogates

**c** Brannin-Fav

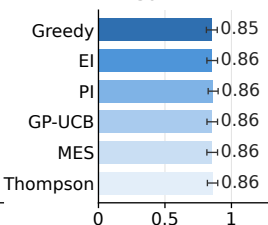
B = 50

**d** Brannin-Unfav

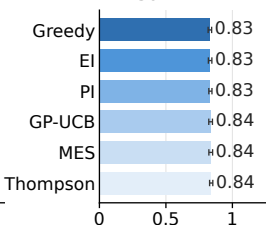
B = 50

**e** Park-Fav

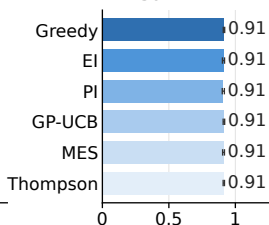
B = 50

**f** Park-Unfav

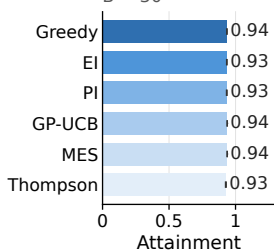
B = 50

**g** COFs

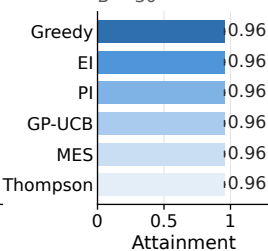
B = 30

**h** FreeSolv

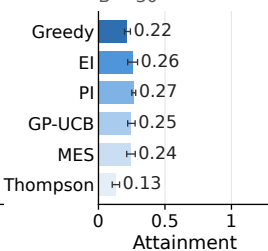
B = 30

**i** Polarizability

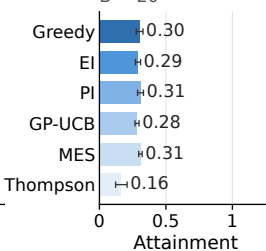
B = 30

**j** HOPV15

B = 30

**k** Matbench-gap

B = 20

c-k: mean $\pm$ s.e. over the five TL surrogates of each surrogate's 20-seed mean attainment (seeds 42-61).

a, b: paired per surrogate (acquisition - greedy), then the share of positive differences (a) or the mean difference (b).

TL: TL-base = base transfer-learning surrogate · TL-PtJ = pretrain-then-joint · TL-E2E = end-to-end joint

TL-SPS = soft parameter sharing · TL-MMD = domain adaptation (MMD)

GP: GP-base = autoregressive multi-fidelity GP (MFGP) · GP-SV = sparse variational MFGP · GP-DKL = deep-kernel GP